

Association of Pain and Steps Per Day in Persons With Mild-to-Moderate, Symptomatic Knee Osteoarthritis: A Mixed-Effects Models Analysis of Multiple Measurements Over Three Years

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Objective. Pain is a consistently reported barrier to physical activity by persons with knee osteoarthritis (OA). Nonetheless, few studies of knee OA have investigated the association of pain with daily walking levels. The current study assessed the relationship of 2 distinct measures of knee pain with objectively measured physical activity in adults with knee OA.

Methods. This was a longitudinal, observational investigation of 59 individuals (48 women; mean $\pm$ SD age 61.1 $\pm$ 6.4 years, mean $\pm$ SD body mass index 28.1 $\pm$ 5.6 kg/m²) with clinical knee OA. Data were collected every 3 months for up to 3 years. Physical activity was characterized as the average steps per day taken over at least 3 days, measured by accelerometry. Pain was measured using 2 patient-administered questionnaires: the pain subscale of the Knee Injury and Osteoarthritis Outcome Score (KOOS-pain) and the P4 pain scale (P4-pain). Mixed-effects models determined the association between pain and physical activity levels (over covariates) among adults with knee OA ($\alpha = 0.05$).

Results. All covariates (age [$\beta = -3.65$, $P < 0.001$], body mass index [$\beta = -3.06$, $P < 0.001$], season [spring/fall $\beta = -6.91$, $P = 0.002$; winter $\beta = -14.92$, $P < 0.001$]) were predictors of physical activity. Neither the inverted KOOS-pain ($\beta = 0.04$, $P = 0.717$) nor P4-pain ($\beta = -0.37$, $P = 0.264$) was associated with physical activity.

Conclusion. Knee pain is not associated with daily walking levels in persons with mild-to-moderate, symptomatic knee OA. While pain management remains an important target of interventions, strategies to increase steps per day in this population should focus on overcoming potentially more crucial barriers to activity participation.

INTRODUCTION

Knee osteoarthritis (OA) is the leading cause of pain, restricted mobility, and disability in older adults (1). Physical activity is recommended for the management of knee OA because it can effectively improve symptoms, including joint pain, as well as physical function and quality of life (2,3). Despite the benefits of physical activity, persons with knee OA engage in less physical activity compared to their healthy counterparts (4–6) and fail to achieve recommended public health guidelines (7,8). Physical activity participation in this clinical population is influenced by numerous factors

(e.g., personal, social, bodily, situational, and environmental) (9,10). Of particular interest is disease-related pain, which is one of the most consistently cited barriers to physical activity in knee OA (9,10).

The question of whether pain is associated with physical activity in knee OA has been the focus of ample work, including systematic reviews (11,12). Nonetheless, the link between pain and physical activity remains ambiguous (11–13). Inconsistencies in study findings are likely attributable to the heterogeneity in the methods used to measure physical activity. To reduce self-report biases, the use of objective measures such as accelerometry is recommended (14,15).

Supported by the CIHR (Joint Motion Program grant to Dr. Brisson, operating grant 102643 to Dr. Maly, New Investigator award to Dr. Maly), the Ontario Ministry of Training, Colleges and Universities (Ontario Graduate scholarships to Dr. Brisson and Mr. Gatti), the Arthritis Society (doctoral salary award to Mr. Gatti), the Canadian Foundation for Innovation (Leaders Opportunity Fund award 27501 to Dr. Maly), and the Ministry of Research and Innovation (Ontario Research Fund award to Dr. Maly).

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No potential conflicts of interest relevant to this article were reported.

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Submitted for publication June 26, 2018; accepted in revised form January 29, 2019.

SIGNIFICANCE & INNOVATIONS

- Pain is a consistently reported barrier to physical activity in knee osteoarthritis (OA); yet, the link between pain and physical activity remains ambiguous.
- This longitudinal, repeated-measures study examined the relationship between pain measurements and steps per day (measured with accelerometry).
- Mixed-effects models, which allow analysis of multiple correlated measurements per participant over time, were applied.
- Knee pain was not associated with daily walking levels in mild-to-moderate, symptomatic knee OA.

Objectively-measured physical activity has been characterized in various ways, including activity frequency (e.g., steps/day) (5,6,13,16–18) and intensity (e.g., light, moderate, vigorous) (7,8,19). Few knee OA studies have formally investigated the association of pain with physical activity, measured objectively as steps per day (13,17,18). Cross-sectionally, pain intensity during activities of daily living was not correlated with steps per day in 160 sedentary overweight/obese adults (17). Only 1 known longitudinal study has directly assessed the relationship between pain and steps per day (13). No association was found between consistent frequent knee pain at baseline and worsening knee pain over 2 years and change in physical activity over 2 years in 1,318 older community-dwelling adults at risk of or with radiographic evidence of knee OA (13).

In order to gain a better understanding of the relationship between pain and physical activity in knee OA, it is essential to account concurrently for the intraindividual and interindividual relationships between pain and physical activity, which can only be accomplished using repeated measures. Furthermore, few longitudinal studies have explored the associations between pain and physical activity. In order to properly analyze multiple correlated measurements per participant over time and deal with missing values, it is imperative that appropriate analyses, such as mixed-effects models, be applied (20). Ascertaining whether pain that is experienced by persons with knee OA is actually associated with reduced walking levels has important implications for disease management.

The primary purpose of this study was to determine the relationship of 2 distinct measures of knee pain with objectively measured physical activity (i.e., steps/day) in adults with clinical knee OA using data acquired every 3 months for up to 3 years. One instrument captured knee pain levels during specific activities over the week prior to physical activity measurements; the other captured diurnal and general activity-related knee pain levels during days that physical activity was recorded. It was hypothesized that by assessing repeated measurements, pain measured using either instrument would be inversely associated with physical activity. A secondary

objective was to explore the relationship between the 2 knee pain measures to ascertain whether these demonstrated convergent construct validity.

PATIENTS AND METHODS

Participants. A convenience sample of consecutive individuals satisfying the eligibility criteria was recruited locally from rheumatology and orthopedic surgery offices. Eligibility criteria included being 40–70 years old and diagnosed with clinical knee OA according to the American College of Rheumatology specifications (21). During recruitment, potential participants were excluded for the following reasons: if they required the use of an adaptive walking aid; if they reported other types of arthritis, a history of lower extremity injury and/or surgery, or ipsilateral hip or ankle conditions; if they had experienced lower extremity trauma or used intraarticular therapies within the previous 3 months; or if they were unable to ascend and descend a 9-step staircase twice consecutively. In cases where participants had bilateral knee OA, the most symptomatic knee was appointed as the study leg. In total, 64 participants met the inclusion/exclusion criteria and were enrolled. All participants provided written, informed consent prior to participating.

Descriptive characteristics of the participants were recorded, including sex, age, height, body mass, and body mass index. Additional data were collected at baseline to characterize clinical and structural disease severities. We used the Knee Injury and Osteoarthritis Outcome Score (KOOS) to assess patients' perceptions of their knee OA (22). The Six-Minute Walk Test, a submaximal task that evaluates walking capacity over long distances, was used to describe functional capacity (23,24). In order to characterize radiographic OA severity at baseline, Kellgren/Lawrence scores (25,26) were derived by an experienced radiologist from digital anteroposterior, weight-bearing, knee radiographs acquired in a standardized fixed-flexion position (27).

This longitudinal, observational study was approved by the Hamilton Integrated Research Ethics Board at McMaster University, Canada. Data were collected from this prospective cohort at 3-month intervals over a maximum of 3 years.

Physical activity. Physical activity was measured using a GT3X+ accelerometer (ActiGraph). Participants were instructed to wear the accelerometer, which was fixed to an adjustable belt, around their waist and aligned with the antero-lateral aspect of the study leg. Participants were asked to wear the accelerometer for 7 consecutive days during waking hours, except during water activities. Vertical accelerations from body movements were sampled at 30 Hz and accumulated over 60-second epochs to yield step counts (i.e., number of steps taken). Accelerometer wear time and number of steps per day were calculated using ActiLife software, version 6 (ActiGraph). Only days during which the accelerometer was worn for at

least 10 hours were considered valid and retained for further analysis (28). Physical activity was characterized as the average number of total steps per day (from both legs) taken over at least 3 days (and up to 7 days) with complete data (29,30). Accelerometer data captured over 3 days yielded reliable estimates of daily physical activity in older adults (intraclass correlation coefficient [ICC] 0.80) (29).

Pain measures. Pain was measured using 2 patient-administered questionnaires: the pain subscale of the KOOS (KOOS-pain) (22) and the P4 pain scale (P4-pain) (31,32). The KOOS-pain is a disease-specific instrument recommended for the evaluation of pain in knee OA (33,34). The KOOS-pain contains 9, 5-level items about knee pain intensity ($n = 8$) and frequency ($n = 1$) during daily activities over the previous week. Each item is rated and converted into scores from 0 (no problem) to 4 (extreme problem). The KOOS-pain was reported as the inverted normalized mean score (of 100), with higher scores indicating more pain (22). Inverted KOOS-pain scores were used to facilitate the interpretation of results by aligning the direction of scores from the KOOS-pain and P4-pain. The KOOS-pain was administered the day before the patient wore the accelerometer, thus reflecting pain levels during the week prior to the physical activity measurements. Data from the KOOS-pain demonstrated the following measurement properties in persons with knee OA: internal consistency (pooled Cronbach's $\alpha = 0.84$), test-retest reliability (pooled ICC 0.90), and absolute measurement error (standard error of measurement = 9.5) (35).

The P4-pain is a generic pain measure developed for musculoskeletal conditions. The P4-pain includes 4 items about knee pain intensity in the morning, afternoon, evening, and with activity over the previous 2 days. Each item is scored from 0 (no pain) to 10 (pain as bad as it can be) on an 11-point numerical pain rating scale, and the scores of all the items are tallied (of 40), with higher scores indicating more pain (31,32). The P4-pain was recorded on 4 occasions (every 2 days over 8 days) and analyzed as the average of 4 measurements taken concurrently during days that physical activity was recorded. Data from the P4-pain showed the following measurement properties in patients with knee OA: internal consistency (Cronbach's $\alpha = 0.91$), test-retest reliability (ICC 0.72), and absolute measurement error (standard error of measurement = 4.5) (36).

Statistical analysis. Mixed-effects models were employed because they allow analysis of repeated measurements on the same participants without violating assumptions of independence necessary for linear regression. To answer the primary research question, 3 models were created and compared. Each model included physical activity (i.e., steps/day) as the dependent variable. The first model included only theoretical covariates as predictors of physical activity: age,

body mass index (BMI), and meteorologic season. Older age (4,6), higher BMI (4,6), and winter season (37,38) are associated with reduced physical activity levels. Meteorologic seasons were defined as follows: summer (June to August), spring/fall (March to May/September to November), and winter (December to February). Spring and fall were combined because meteorologic conditions during these months are comparable in Canada. The predictor season was dummy-coded using summer as the reference season, thus comparing how physical activity during spring/fall and winter was different from summer. The second and third models were generated by adding the KOOS-pain and the P4-pain, respectively, to the covariate model. Each model was fit using maximum likelihood estimation. The overall predictive ability of the KOOS-pain and P4-pain models was compared to the covariate model using the likelihood ratio test. All tests used a statistical significance threshold set to an alpha level of 0.05. Assumptions, including normality of residuals and homogeneity of variance, were assessed. To satisfy assumptions, all models used log-transformed physical activity as the dependent variable as well as a random intercept grouped by participant. Random intercept and random slope models were assessed; only the random intercept model significantly improved model fit (likelihood ratio test $P < 0.001$) and model assumptions (normality of residuals). All analyses were performed in Stata, version 13.1. To facilitate interpretation of the log-transformed data, the presented β coefficients were transformed ($[\exp(\beta) - 1] \times 100$) and represent the percentage change in the dependent variable for a 1-unit change in the predictor.

To answer the secondary research question, a mixed-effects model was employed. The model included KOOS-pain

Table 1. Descriptive statistics of 59 participants at baseline (48 women)*

Variable	Mean $\pm$ SD	Range
Age, years	61.1 $\pm$ 6.4	41–69
Height, meters	1.63 $\pm$ 0.09	1.46–1.94
Body mass, kg	75.0 $\pm$ 16.1	49.4–117.0
Body mass index, kg/m ²	28.1 $\pm$ 5.6	19.7–41.8
Other symptoms†	24.5 $\pm$ 15.8	0–75
Function in daily living†	18.1 $\pm$ 15.4	0–59
Function in sports and recreation, $n = 57$ †	31.0 $\pm$ 22.9	0–85
Knee-related quality of life, $n = 58$ †	35.4 $\pm$ 18.7	0–87
Six-Minute Walk Test, meters	528.4 $\pm$ 99.3	246–771
KOOS-pain‡	24.6 $\pm$ 17.1	0–62
P4-pain‡	6.4 $\pm$ 6.6	0–25
Physical activity, steps/day	7,158 $\pm$ 3,071	1,274–20,895

* KOOS-pain = Knee Injury and Osteoarthritis Outcome Score pain subscale; P4-pain = P4 pain scale.

† Measured with subscales of the KOOS (22). Inverted KOOS scores were used. Scores range 0–100, where higher scores reflect worse symptoms and function.

‡ Measured with the P4 pain scale (31). Scores range 0–40, where higher scores reflect more pain.

as the dependent variable and P4-pain as the independent variable. It also utilized a random intercept grouped by participant to achieve assumptions of normality and homoscedasticity. To provide an indicator of convergent construct validity, the coefficient of determination (R^2) for the fixed portions of the fitted model was calculated ($R^2 = 1 - [\text{sum of squares}_{\text{residual}} / \text{sum of squares}_{\text{total}}]$).

RESULTS

Data from 59 participants (of 64 participants who enrolled) were included in the current analysis. Only participants with valid/complete data for all 3 measures of interest (i.e., KOOS-pain, P4-pain, physical activity) for at least 2 time points (i.e., repeated-measures) were retained. Data from 5 of the 64

participants who enrolled were excluded entirely from the analysis due to missing data (see Supplementary Figure 1, available on the *Arthritis Care & Research* web site at <http://onlinelibrary.wiley.com/doi/10.1002/acr.23842/abstract>). The final sample used for analysis comprised 11 men and 48 women, with the following baseline Kellgren/Lawrence scores: grade I = 2; II = 23; III = 20; IV = 14. Additional baseline descriptive statistics are given in Table 1. All models included 513 observations from 59 participants collected over a maximum of 3 years. The number of observations per participant ranged from 2 to 13, with a mean $\pm$ SD 8.7 ± 3.5 observations. In order to demonstrate the intraparticipant and interparticipant variability in outcome measures over the course of the study, average measurements (with 95% confidence intervals for the measurements) for physical activity, KOOS-pain, and P4-pain are plotted for each participant

Table 2. Associations of covariates (age, BMI, season) and potential predictors (KOOS-pain, P4-pain) with the dependent variable (physical activity measured as steps per day)*

Model	Predictor	β coefficient	Variance	P
Covariates				
Log likelihood = -18.88; Wald $\chi^2(4) = 95.41$; $P < 0.001$				
	Age, years	-3.65		<0.001†
	BMI	-3.06		<0.001†
	Spring/fall	-6.91		0.002†
	Winter	-14.92		<0.001†
	Random intercept		0.154	
	Residual		0.043	
KOOS-pain				
Log likelihood = -18.81; Wald $\chi^2(5) = 95.57$; $P < 0.001$				
	Age, years	-3.68		<0.001†
	BMI	-3.10		<0.001†
	Spring/fall	-6.89		0.003†
	Winter	-14.88		<0.001†
	KOOS-pain (22)‡	0.04		0.717
	Random intercept		0.155	
	Residual		0.043	
P4-pain				
Log likelihood = -18.25; Wald $\chi^2(5) = 96.89$; $P < 0.001$				
	Age, years	-3.54		<0.001†
	BMI	-2.90		<0.001†
	Spring/fall	-6.93		0.002†
	Winter	-15.16		<0.001†
	P4-pain (31)§	-0.37		0.264
	Random intercept		0.151	
	Residual		0.043	

* All models used log-transformed physical activity as the dependent variable and a random intercept grouped by participant. To facilitate interpretation, β coefficients were transformed ($[\exp(\beta) - 1] \times 100$) and represent the percentage change in the dependent variable for a 1-unit change in the predictor. All models included 59 groups (participants) and 513 observations. Summer was used as the reference season. Seasons were defined as summer (June–August), spring/fall (March–May/September–November), and winter (December–February). BMI = body mass index; KOOS-pain = Knee Injury and Osteoarthritis Outcome Score pain subscale; P4-pain = P4 pain scale.

† Significant ($P < 0.05$).

‡ Inverted KOOS-pain scores were used. Scores range 0–100, where higher scores reflect more pain.

§ Scores range 0–40, where higher scores reflect more pain.

(see Supplementary Figures 2, 3, and 4 at <http://onlinelibrary.wiley.com/doi/10.1002/acr.23842/abstract>).

Association between pain and physical activity. The covariate model revealed that age ($\beta = -3.65$, $P < 0.001$) (see Supplementary Figure 5, available on the *Arthritis Care & Research* web site at <http://onlinelibrary.wiley.com/doi/10.1002/acr.23842/abstract>) and BMI ($\beta = -3.06$, $P < 0.001$) (see Supplementary Figure 6 at <http://onlinelibrary.wiley.com/doi/10.1002/acr.23842/abstract>) were inversely related to physical activity (Table 2). The seasons of spring/fall ($\beta = -6.91$, $P = 0.002$) and winter ($\beta = -14.92$, $P < 0.001$) were associated with lower levels of physical activity compared to summer, with winter having the least activity (see Supplementary Figure 7, at <http://onlinelibrary.wiley.com/doi/10.1002/acr.23842/abstract>). The second model, which included the inverted KOOS-pain ($\beta = 0.04$, $P = 0.717$) (Figure 1), did not differ significantly from the covariate model (likelihood ratio test $\chi^2[1] = 0.13$, $P = 0.718$). The third model, which included P4-pain ($\beta = -0.37$, $P = 0.264$) (Figure 2), did not differ significantly from the covariate model (likelihood ratio test $\chi^2[1] = 1.24$, $P = 0.266$).

Association between KOOS-pain and P4-pain. While visualizing scatterplots and residual-versus-fitted plots for a linear fit between KOOS-pain and P4-pain, it was determined that the relationship was nonlinear; thus, a squared term ($P4\text{-pain}^2$) was included. The model investigating the relationship between the 2 pain measures revealed that P4-pain ($\beta = 2.370$, $P < 0.001$) and $P4\text{-pain}^2$ ($\beta = -0.042$, $P < 0.001$) were significant predictors of

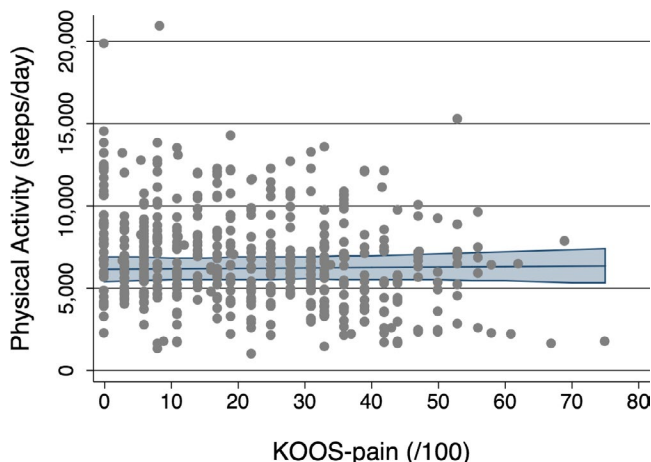


Figure 1. Fitted physical activity (back-transformed) as a function of the Knee Injury and Osteoarthritis Outcome Score (KOOS-pain) and the coinciding 95% confidence interval band. Individual measurements are plotted in gray ($n = 513$). Physical activity was characterized as the average number of total steps per day (from both legs) taken over at least 3 days (and up to 7 days), measured by an accelerometer. The inverted KOOS-pain scores were reported of 100, with higher scores indicating more pain. KOOS-pain scores reflected knee pain levels during daily activities over the week prior to the physical activity measurements (22).

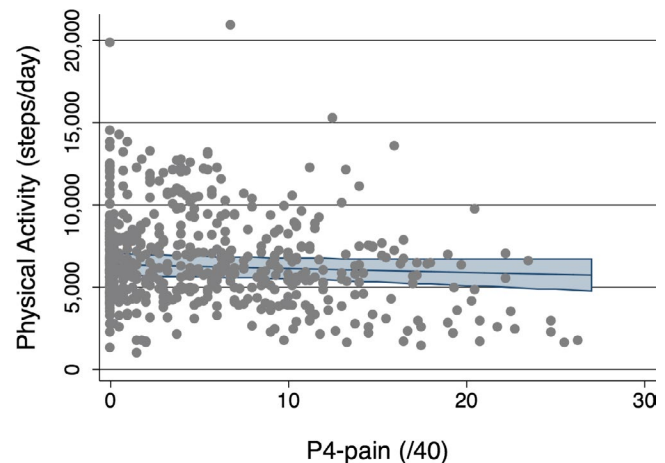


Figure 2. Fitted physical activity (back-transformed) as a function of the P4 pain scale (P4-pain) and the coinciding 95% confidence interval band. Individual measurements are plotted in gray ($n = 513$). Physical activity was characterized as the average number of total steps per day (from both legs) taken over at least 3 days (and up to 7 days), measured by an accelerometer. P4-pain scores were reported of 40, with higher scores indicating more pain (31). P4-pain scores reflected a 4-measurement average of diurnal and activity-related knee pain levels over 8 days, measured concurrently with physical activity measurements.

the inverted KOOS-pain (Table 3) (Figure 3). The fixed effects of the fitted model yielded an R^2 of 0.579, demonstrating that these measures possessed high convergent validity.

DISCUSSION

In the current study, disease-related knee pain was not associated with steps per day in persons with mild-to-moderate, symptomatic knee OA. This study sought to improve our understanding of the complex relationship between pain severity and physical activity habits in knee OA. The novel contribution and primary strength of this study were the use of multiple measurements to account for potential intraindividual and interindividual relationships, which allowed us to ascertain whether within- and between-participant fluctuations in pain are related to physical activity levels. Results suggest that daily walking levels observed in persons with knee OA are not explained by knee pain severity, emphasizing the notion that pain is not a primary barrier to physical activity participation in this population. Therefore, while pain management remains an important target of clinical interventions, pain alleviation is not likely to increase physical activity levels. Strategies aiming to increase steps per day in knee OA should focus on overcoming potentially more crucial barriers to activity participation, such as lack of knowledge and motivation as well as overall sedentary lifestyle.

Some prior evidence supports the absence of a relationship between knee pain and steps per day in persons with knee OA (13,17). Along with the current investigation, these studies utilized

Table 3. Relationship between P4-pain and KOOS-pain measures*

Model	Predictor	β coefficient	Variance	SE	P
P4-pain and KOOS-pain Log likelihood = -1,799.44; Wald $\chi^2(2)=237.11$; $P < 0.001$					
	Intercept	11.928		1.429	<0.001†
	P4-pain	2.370		0.011	<0.001†
	P4-pain ²	-0.042		0.011	<0.001†
	Random intercept		65.191	14.846	
	Residual		49.290	3.320	

* A squared term (P4-pain²) was included to model the nonlinear relationship visualized between measures. The model included 59 groups (participants), 513 observations, and a random intercept grouped by participant. Values are P4-pain (31) and the inverted score from the pain subscale of the KOOS-pain (Knee Injury and Osteoarthritis Outcome Score pain subscale) (22). For both the P4-pain and KOOS-pain, higher scores reflect more pain. P4-pain = P4 pain scale.

† Significant ($P < 0.05$).

objectively measured steps per day as a measure of physical activity, and they measured pain using standardized, validated instruments. In contrast, only 1 known study (cross-sectional) reported an association between daily walking levels and pain (18). It should be noted that this association was only found for specific activities and subgroups, and pain was analyzed using single-item scores rather than the recommended subscale total scores (18). Caution should be taken in interpreting these findings because testing individual items in multiple subgroups inflates the Type I error rate. While it is conceivable that daily walking levels may be associated with activity-specific pain intensity depending on disease severity, such speculations remain to be examined longitudinally using standardized, validated instruments.

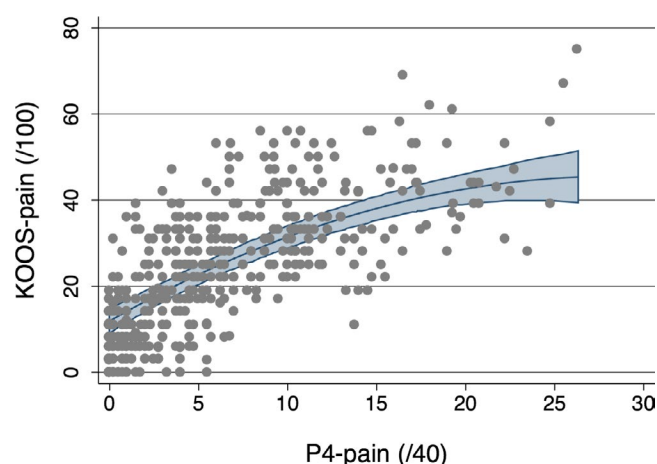


Figure 3. Fitted Knee Injury and Osteoarthritis Outcome Score (KOOS-pain) as a function of the P4 pain scale (P4-pain) and the coinciding 95% confidence interval band. Individual measurements are plotted in gray ($n = 513$). The inverted KOOS-pain scores were reported of 100, with higher scores indicating more pain. KOOS-pain scores reflected knee pain levels during daily activities over the week prior to the physical activity measurements (22). P4-pain scores were reported of 40, with higher scores indicating more pain (31). P4-pain scores reflected a 4-measurement average of diurnal and activity-related knee pain levels over 8 days, measured concurrently with physical activity measurements. Color figure can be viewed in the online issue, which is available at <http://onlinelibrary.wiley.com/doi/10.1002/acr.23842/abstract>.

The current study expands on previous investigations by evaluating pain using 2 distinct, validated and reliable instruments: the 9-item KOOS-pain, which reflected knee pain levels during specific activities over the week prior to physical activity measurements; and the P4-pain scale, which reflected diurnal and general activity-related knee pain levels during days that physical activity was recorded. Considered collectively, findings from the present and past studies provide robust evidence that knee pain severity is not related to steps per day in knee OA. Specifically, findings persisted regardless of the instruments used to assess pain, the interval over which pain was recorded relative to physical activity, and whether pain was captured during activities of daily living, at different times of day, or with general activity. It should be noted, however, that results from the current study were observed in a sample of individuals who exhibited relatively mild disease-related symptoms; mean $\pm$ SD pain scores at baseline were 24.6 ± 17.1 (of 100) for the inverted KOOS-pain and 6.4 ± 6.6 (of 40) for P4-pain. Pain scores were not discretely reported in the other longitudinal work (13), while the sample from the cross-sectional study also displayed relatively low pain levels (mean $\pm$ SD 5.8 ± 2.8 [of 20] for the Western Ontario and McMaster Universities Osteoarthritis Index pain subscale [WOMAC-pain]) (17). It may be that pain intensity in these investigations was not high enough to have an impact on daily walking levels. Conversely, it may be that these individuals were not active enough for their physical activity habits to be altered significantly due to knee pain.

Despite the many benefits of regular physical activity to the management of knee OA, many persons with this condition fail to achieve recommended levels (7,8). This fact reinforces the notion that physical activity should be at the core of treatment strategies for knee OA. It is critical to identify factors that can hinder daily walking behaviors in at-risk populations. Findings from this study help to advance this goal by confirming that knee pain levels (a factor popularly believed to be a primary cause of reduced physical activity in knee OA) are not associated with intraindividual or interindividual fluctuations in walking levels. Accordingly, strategies aiming to increase physical activity in this patient group must overcome other potential key barriers (9,39,40), possibly by tailoring interventions to individual patient preferences and needs

and by providing access to proven supplemental resources (e.g., patient advice/education, arthritis self-management programs, incentives) (41–43).

A secondary objective of this work was to explore the relationship between the KOOS-pain and the P4-pain measurements in this sample. Results from the mixed effects model demonstrated a slight curvilinear relationship between the 2 measures ($R^2 = 0.579$), indicating stronger agreement than that reported between the P4-pain and WOMAC-pain ($r = 0.67$, $R^2 = 0.449$) (44). Imperfect agreement between the P4-pain and KOOS-pain measures (i.e., $R^2 < 1.0$) is to be expected for several reasons. First, each instrument assessed pain over a different timeframe (i.e., the week preceding and the week corresponding to physical activity measurements). Second, the P4-pain asked about pain at different times of day (i.e., morning, afternoon, evening) and with activity, while the KOOS-pain asked about pain with specific activities of daily living. A strong but imperfect agreement suggests that these measures effectively assess the construct of pain but also that multiple measures should be used when conducting analyses aiming to capture all relevant themes of pain.

Limitations of this study should be acknowledged. The study design precluded the establishment of directionality between change in pain and physical activity. That is, pain may have already altered activity participation, and vice versa, prior to the study. Only pain in the study leg was evaluated; pain in the contralateral leg of participants with bilateral knee OA was not controlled for in the analyses. No insight was gained into the relationship between knee pain and other measures of activity (e.g., intensity, type). In addition, 2 distinct pain measures were used in this study; nonetheless, neither one discriminated between types of pain (i.e., constant, intermittent), and only 1 item about pain frequency (KOOS-pain) was included. Future work should examine the relationship between physical activity and different pain domains. Furthermore, while the mixed-effects models adjusted for important covariates, including age, BMI, and season, they did not control for additional variables that could have potentially influenced physical activity levels, such as self-efficacy (45), depressive symptoms (13), fatigue (46), pain-related activity interference (47), and education status (48). Finally, although the sample in the current study was selected based on clinical knee OA criteria, participants exhibited relatively high function and low levels of pain and other symptoms. Thus, results can only be generalized to persons with matching characteristics.

In conclusion, knee pain severity is not associated with physical activity levels, measured objectively as steps per day, in persons with mild-to-moderate, symptomatic knee OA. Walking levels in individuals with knee OA are likely influenced more strongly by factors other than disease-related pain. Interventions aiming to augment daily walking levels in this clinical population should focus on overcoming key barriers to physical activity

participation and adherence while conjointly attending to pain and symptom management.

ACKNOWLEDGMENTS

The authors thank Christine Dibblee and Emily Wiebenga from the Mobilize Laboratory for their help with data collection, as well as the study participants for making this research possible.

AUTHOR CONTRIBUTIONS

All authors were involved in drafting the article or revising it critically for important intellectual content, and all authors approved the final version to be submitted for publication. Dr. Maly had full access to all of the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis.

Study conception and design. Maly.

Acquisition of data. Brisson, Maly.

Analysis and interpretation of data. Brisson, Gatti, Maly.

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